Real-Time Multimodal Driver Stress Monitoring and Adaptive Feedback via Biosignals and Conversational AI

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***Abstract*— Driver stress is a critical factor that influences road safety, with elevated stress levels often leading to impaired decision-making, delayed responses, and higher risk of accidents. Traditional monitoring approaches based on visual cues or unimodal physiological signals lack robustness in real-world conditions. This work proposes a multimodal stress detection framework that integrates electrocardiogram (ECG), electromyogram (EMG), respiration (RESP), heart rate (HR) and galvanic skin response (GSR) signals collected from 17 drivers over 22 hours of naturalistic driving. The pipeline consists of signal preprocessing, multimodal feature extraction, and stress state labeling based on task intensity. Baseline classification employs Fisher Discriminant Analysis (FDA), k-Nearest Neighbor (kNN), Random Forest (RF) and stacking ensemble methods, with both within-driver and cross-driver validation. The system is designed for extensibility to deep learning models such as multimodal CNN-LSTMs and Transformers, fuzzy logic integration for uncertainty handling, and real-time conversational feedback for driver support. Performance will be validated using accuracy, F1 score, and generalization metrics of the driver. The proposed system contributes to the development of robust, multimodal, and human-centered stress recognition technologies for intelligent transportation systems.**

***Index Terms*— Driver Stress Detection, Multimodal Signals, ECG, GSR, Feature Extraction, Ensemble Learning, Fuzzy Logic**

I. INTRODUCTION

Driver stress is a critical factor influencing road safety, cognitive performance, and overall well-being. Elevated stress levels can impair decision-making, slow reaction times, and increase the likelihood of traffic accidents. Traditional driver monitoring systems primarily rely on visual cues, such as facial expressions or eye tracking, or on single physiological measurements, including heart rate or galvanic skin response. However, these methods are often limited in real-world conditions due to environmental noise, motion artifacts, inter-individual variability, and deliberate masking behaviors, resulting in inconsistent or unreliable stress assessment.

Stress is inherently a multidimensional physiological response involving interactions among the cardiovascular, muscular, respiratory, and electrodermal systems. Capturing this complexity requires simultaneous monitoring of multiple physiological modalities. Focusing on a single signal modality often fails to represent the full spectrum of stress responses, potentially leading to suboptimal detection performance and poor generalizability across different individuals.

To address these limitations, this work proposes a multimodal physiological framework for driver stress detection. The system integrates synchronized recordings of electrocardiogram (ECG), electromyogram (EMG), respiration (RESP), heart rate (HR), and galvanic skin response (GSR) from both hand and foot sensors. By leveraging multimodal fusion, the system exploits complementary information across different physiological systems, enhances robustness against artifacts, and improves the accuracy and reliability of stress classification.

This research makes several contributions. First, we develop a multimodal dataset of 17 drivers with approximately 22 hours of synchronized physiological recordings, processed through preprocessing, resampling, segmentation, and modality-specific feature extraction. Second, stress levels are annotated based on task intensity (low, medium, high), providing structured supervision for model training and enabling both within-driver and cross-driver evaluation. Third, we implement multiple supervised learning approaches, including Fisher Discriminant Analysis (FDA), k-Nearest Neighbor (kNN), Random Forest (RF), and stacking ensembles, establishing robust baseline performance. Finally, the proposed methodology lays the groundwork for future extensions, such as advanced deep learning models (multimodal CNN-LSTMs and Transformers), fuzzy logic handling of inter-individual variability, and integration with conversational AI for real-time stress interventions.

The remainder of this paper is structured as follows: Section II presents a literature survey on driver stress detection using physiological signals. Section III details the dataset construction, feature extraction, and annotation process. Section IV describes the implementation of baseline and ensemble classification models and presents the experimental results. Section V discusses potential future extensions and applications, followed by conclusions and references.

II. LITERATURE SURVEY

A. Background

Driver stress is a critical factor influencing road safety, cognitive performance, and overall well-being. Elevated stress levels can impair decision-making, slow reaction times, and increase accident risk. Traditional driver monitoring approaches rely on visual cues or single physiological signals, which are often unreliable in real-world conditions due to noise, motion artifacts, individual variability, and deliberate masking behaviors. Stress, however, is a multidimensional physiological response involving cardiovascular, muscular, respiratory, and electrodermal systems. Therefore, robust detection requires monitoring multiple physiological modalities simultaneously.

B. Literature Survey

Healey and Picard [**?**] investigated whether physiological sensors could detect driver stress in real-world driving. Using ECG, EMG, respiration, and skin conductance signals from 24 drives (50 min each), they achieved 97% accuracy for low, medium, and high stress segments. Their study highlighted the potential of physiological monitoring for adaptive in-vehicle systems but lacked real-time inference and structured segment-level stress labeling.

Vargas-Lopez et al. [**?**] focused on stress detection using EMG signals with statistical time features and explainable SVM classifiers. They demonstrated high predictive performance (AUC 0.97) in controlled driving-like conditions, emphasizing EMG robustness against skin moisture and cardiac variability. Limitations included single-modality data, limited applicability to dynamic real-world conditions, and untested real-time performance.

Healey et al. [**?**] extended their work to wearable and car-mounted multimodal affect recognition systems. With 36 hours of data from 10 drivers across lab, ambulatory, and automotive scenarios, they achieved stress detection rates of 96% (task-based) and 89% (questionnaire-based). This study underscored the importance of multimodal fusion but faced challenges with motion noise, individual variability, and sensor durability.

Lopez-Martinez et al. [**?**] proposed a multi-view, multi-task learning approach using EDA and heart rate to classify high and low stress in real-world driving. Using MIT drivedb, HciLab, and AffectiveROAD datasets (total 40 drives), the MT-MKL model improved performance over single-task approaches, with best accuracies of 93%, 83%, and 71%. Their findings emphasized personalization in stress recognition but did not explore moderate stress levels and lacked multimodal inputs like facial cues or speech.

Zontone et al. [**?**] analyzed skin potential response (SPR) and ECG in simulated manual and autonomous driving. Using SVM classifiers on SPR features from 13 volunteers, they achieved 73% average accuracy. Manual driving induced more frequent physiological stress responses, whereas autonomous driving elicited fewer but higher-intensity peaks. Limitations included small sample size, fixed stress labels, and exclusion of ECG for classification due to variability.

Chen et al. [**?**] developed a multimodal driving stress detection system using ECG, GSR, and respiration. Features were extracted from time, frequency, and wavelet domains, with PCA and Sparse Bayesian Learning for dimensionality reduction, and kernel classifiers for stress evaluation. Experiments on 14 drivers in real-world conditions yielded over 99% drive-level classification accuracy and 89% cross-drive validation accuracy. The study demonstrated the value of multi-view data fusion but noted challenges in generalizing to diverse populations and continuous real-time deployment.

III. PROPOSED METHODOLOGY

The proposed framework for multimodal driver stress detection leverages physiological signals from multiple sources, integrates them into a unified feature set, and performs robust stress classification. The methodology consists of seven well-defined stages described below.

A. Data Acquisition

The study utilizes a multimodal physiological dataset collected from 17 drivers, totaling approximately 22 hours of recordings. Each driving session lasts between 1 and 1.3 hours and includes synchronized signals from electrocardiogram (ECG), electromyogram (EMG), respiration (RESP), heart rate (HR), and galvanic skin response (GSR) from both hand and foot sensors. Sampling frequencies vary across modalities, ranging from 15 Hz to 31 Hz depending on the sensor type. The dataset captures naturalistic driving conditions, including baseline, moderate, and high-stress scenarios.

B. Preprocessing

To ensure signal quality and uniformity, each modality undergoes noise reduction, synchronization, and segmentation. Preprocessing steps include:

- **ECG:** Band-pass filtering between 0.5–40 Hz to remove baseline wander and high-frequency noise.
- **EMG:** Computation of smoothed absolute values to reduce muscle signal artifacts.
- **Respiration:** Band-pass filtering between 0.1–0.5 Hz to capture breathing dynamics.
- **GSR:** Low-pass filtering with a 1 Hz cutoff to isolate slow conductance changes.

- **Resampling:** All modalities resampled to a uniform 100 Hz to facilitate multimodal fusion.
- **Segmentation:** Signals divided into 5-minute windows to preserve temporal consistency.

C. Feature Extraction

Features are extracted from each modality to quantify physiological responses relevant to stress:

- **ECG/HRV:** Mean heart rate, SDNN, RMSSD, LF/HF ratio, and other time- and frequency-domain HRV metrics.
- **EMG:** Mean amplitude, RMS, skewness, kurtosis, and zero-crossing counts to capture muscular tension.
- **Respiration:** Average breathing rate, inter-breath variability, and spectral power to monitor respiratory patterns.
- **GSR:** Mean conductance, standard deviation, peak counts, and slope to measure sympathetic arousal.

These features collectively form a high-dimensional representation of driver physiological state suitable for supervised classification.

D. Stress State Labeling

Stress labels are assigned based on task intensity and experimental conditions:

- **Low stress:** Drivers 1–5 under baseline driving conditions.
- **Medium stress:** Drivers 6–10 during moderate driving challenges.
- **High stress:** Drivers 11–17 navigating city traffic and demanding scenarios.

This labeling strategy provides structured supervision for classification models, enabling both within-driver and cross-driver evaluation.

E. Output Storage

Extracted features and labels are stored systematically for reproducibility and downstream analysis:

- Features are stored per session in a `features/` directory.
- Corresponding stress annotations are stored in `annotations/`.
- An aggregated dataset `stress_features_all.csv` is generated for global analysis and model development.

F. Baseline Classification

Multiple supervised learning approaches are implemented to establish robust baseline performance:

- **Fisher Discriminant Analysis (FDA):** Linear projection-based classifier for interpretability and dimensionality reduction.
- **k-Nearest Neighbor (kNN):** Non-parametric classifier capturing local feature similarities.
- **Random Forest (RF):** Ensemble of decision trees that improves performance on high-dimensional, noisy features.
- **Stacking Classifier:** Meta-classifier combining FDA, kNN, SVM, RF, and MLP for enhanced predictive power.

Validation is performed at both within-driver and cross-driver levels to evaluate model generalizability and robustness across individuals.

G. Future Extensions

The proposed methodology provides a foundation for advanced developments:

- **Deep Learning:** CNN-LSTM and Transformer-based architectures for temporal and cross-modal stress dynamics.
- **Fuzzy Logic:** Handling uncertainty and inter-individual variability in stress responses.
- **Conversational Feedback:** Integration with lightweight LLMs for real-time stress interventions.

IV. Block Diagram and Workflow

The proposed system for ECG-based driver stress detection follows a structured workflow as illustrated in the block diagram shown in Fig. 1. The framework consists of several sequential stages: data acquisition, preprocessing, feature extraction, labeling, model training, prediction, and feedback generation. Each stage is designed to ensure accurate stress classification and effective real-time intervention.

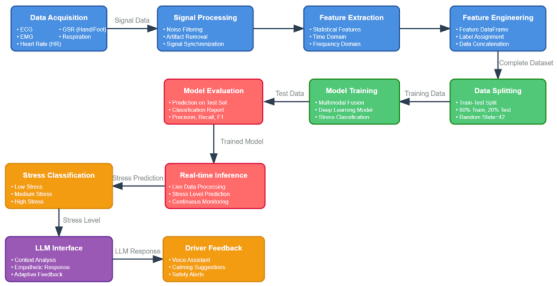

Fig. 1. Block diagram illustrating the workflow of the proposed driver stress detection system.

Workflow Description:

- **Data Acquisition:** Multimodal physiological signals such as ECG, EMG, respiration, and GSR are collected from drivers during simulated or real driving sessions.
- **Preprocessing:** Raw signals are cleaned using filtering techniques to remove noise and artifacts, ensuring quality input for feature extraction.
- **Feature Extraction:** Key temporal and frequency-domain features are computed from the physiological signals to capture stress-relevant patterns.
- **Labeling:** Stress levels are annotated based on predefined rules or expert intervention, forming the ground truth for supervised model training.
- **Model Training:** Machine learning or deep learning models are trained using the labeled features to learn patterns associated with different stress levels.

- **Prediction:** The trained models predict stress levels for new driver data in real-time or offline scenarios.
- **Feedback Generation:** Based on predicted stress levels, adaptive interventions or alerts are generated, which may include calming prompts or real-time notifications to the driver.

V. Algorithms and Technologies Used

- **Fisher Discriminant Analysis (FDA):** Linear projection-based classifier to maximize class separability.
- **k-Nearest Neighbors (kNN):** Non-parametric algorithm leveraging proximity in feature space for classification.
- **Random Forest (RF):** Ensemble of decision trees for handling high-dimensional and noisy features.
- **Stacking Classifier:** Meta-classifier combining FDA, kNN, RF, SVM, and MLP for robust performance.
- **Python Libraries:** `numpy`, `scipy`, `sklearn`, and `pandas` for signal processing, feature extraction, and model implementation.
- **Visualization Tools:** `matplotlib` and `seaborn` for plotting features, confusion matrices, and performance metrics.

VI. Dataset Description

The dataset comprises multimodal physiological recordings from 17 drivers under varying stress conditions:

- **ECG:** Captures cardiac activity at 100 Hz after resampling.
- **EMG:** Monitors muscular activity at 100 Hz.
- **Respiration:** Measures breathing patterns between 0.1–0.5 Hz.
- **GSR (Hand and Foot):** Monitors skin conductance at 15–31 Hz, smoothed and low-pass filtered.
- **Labels:** Low (drivers 1–5), Medium (drivers 6–10), High (drivers 11–17) stress levels.
- **Session Duration:** Each session lasts approximately 1–1.3 hours.

VII. Performance Metrics

To quantitatively evaluate the proposed solution, standard classification metrics are employed. These metrics assess the performance of the model across all classes and provide insights into its predictive capabilities.

Accuracy: Accuracy measures the overall correctness of the classifier and is defined as the ratio of correctly predicted instances to the total number of predictions:

$$Accuracy = \frac{TruePos + TrueNeg}{TruePos + FalsePos + FalseNeg + TrueNeg} \tag{1}$$

Recall (True Positive Rate, TPR): Recall quantifies the ability of the classifier to correctly identify positive instances. It is defined as:

$$Recall = \frac{TruePos}{TruePos + FalseNeg} \tag{2}$$

Precision: Precision evaluates the correctness of predicted positive instances and indicates how many of the predicted positives are actually positive:

$$Precision = \frac{TruePos}{TruePos + FalsePos} \tag{3}$$

F1-score: The F1-score is the harmonic mean of Precision and Recall, providing a balanced evaluation metric that accounts for both false positives and false negatives:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

These metrics collectively offer a comprehensive assessment of model performance, ensuring robustness and reliability in multiclass stress classification scenarios.